Shadow-Less Intrinsic Hyperspectral Point Cloud Generation From HSIs and LiDAR

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***Abstract*— Generating hyperspectral point cloud from hyperspectral images (HSIs) and light detection and ranging (LiDAR) has become more and more common in the remote sensing field and supported various applications. One challenge here is that hyperspectral imaging is a passive imaging method and is suffering from shadows in a natural scene. Intrinsic information recovery can effectively eliminate the spectral variation caused by illumination changes; however, it assumes a uniform light and neglects the shadows in the scene. In this article, we provide a novel hyperspectral point cloud intrinsic model that can detect the shaded regions and recover reflectance information in them. We first estimate the global illumination of the scene using an intrinsic information recovery method. Then, we perform supervoxel segmentation on hyperspectral point cloud to calculate the blocking relation of supervoxels and therefore accurately detect shaded regions. Finally, we estimate the illumination and reflectance of shaded regions based on an illumination-invariant spectral prior. The experimental results show that the proposed method can effectively detect shaded areas and robustly generate shadow-less intrinsic hyperspectral point cloud.**

***Index Terms*— Hyperspectral images (HSIs), hyperspectral point cloud, intrinsic information recovery, light detection and ranging (LiDAR), shadow removal.**

I. Introduction

INTRINSIC image decomposition (IID) problem was first raised in the computer vision field, which is designed to recover the reflectance component from a single red, green, blue (RGB) image by removing shading component [1], [2], [3], [4], since the reflectance component is directly related to intrinsic characteristic of different materials and is invariant to illumination condition, and the shading component usually characterizes the shape and texture of a certain object. The input image is calculated by the multiplication of the two intrinsic components element by element, which is expressed by the following equation:

$$I = sR \tag{1}$$

where I is the input image, s presents the shading image, and R presents the reflectance image. To reduce the complexity, it is common to rewrite this equation into a logarithmic form. However, we all know that the IID problem is a typical ill-posed task, in which case, how to propose appropriate assumptions based on prior knowledge becomes the most critical step. At present, a widely accepted assumption is that the adjacent pixels with similar chromaticity and intensity should have similar reflectance as well [5].

Compared to the "intrinsic image" problem, a more general problem named "shape from shading" was further raised in the computer vision field, which is to separate the shading component into illumination and shape parts. This can be achieved by adopting a new depth prediction method with natural image statistics analysis [6], [7] or convolutional neural network (CNN) [8], [9], and the former usually takes only a single segmented object as input, while the latter deals directly with a complete input image scene. For more general cases, where objects have spatially varying illumination by blocking or supporting each other, additional sources are needed to recover the reflectance, illumination and shape components from the entire image scene, such as a sequence of images under different illumination [7], [10], [11], depth cues obtained by depth sensors [12], [13], [14], [15], or more information on incident light direction [16].

Although there are sorts of valid IID models aimed at RGB and RGB-D images, these works are not suitable for other forms of data, such as HSI or light detection and ranging (LiDAR) point clouds. HSI is a hyperspectral cube with rich spectral information that covers continuous bands containing the visible range and near-infrared range. HSI differs greatly from RGB as the former has only three spectral channels, while the latter may contain hundreds of spectral channels. High spectral dimensionality means that there is high correlation and redundancy between spectral channels of HSI. In addition, remote sensing HSI has low spatial resolution, which causes spectral aliasing problems [17], making it hard for RGB IID models to be extended to HSI. So far, some works have been done on the IID of a single HSI [18], [19], [20], successfully decomposing HSIs into hyperspectral reflectance component and shadow component. Nevertheless, these methods still present the following disadvantages: 1) the mixtures of surface material in a single pixel of HSI are not considered; 2) the shadow effect is usually ignored; and 3) properties, including shape, depth, and illumination, are not recovered. To handle these problems, in this article, we provide a novel shadow-less hyperspectral point cloud joint IID approach for HSI and LiDAR. Instead of generating a 2-D reflectance image and a shading image from HSI, our model will generate a

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3-D intrinsic hyperspectral point cloud (IHSPC) with 3-D reflectance and shape information.

In the remote sensing field, spectral 3-D data have been found useful in various applications, e.g., target detection, biomass estimation, and classification. Also, it is recently a trend to develop hyperspectral LiDAR systems, which can acquire one-shot topography and spectral intensity information [21], [22], [23], [24], [25]. Active hyperspectral imaging has the advantage of no shadows and no registration problems between datasets. However, producing such a system is much more expensive than using separate sensors. As an application of the proposed method, it gives a solution for generating integrated spectral 3-D data from two cheaper sensors.

The combination of 3-D elevation and spectral information provided by LiDAR and HSI, respectively, is of particular interest because there is a strong complementarity between the two data sources [26], [27]. On the one hand, hyperspectral imaging obtains images with rich spectral channels that provide important information for distinguishing different materials on the ground, but the 2-D plane imaging causes 3-D objects to lose geometric structure information. LiDAR, on the other hand, provides intensive 3-D laser scanning of the ground objects and 3-D spatial information for scanned area. However, the lack of spectral features makes LiDAR relatively weak in distinguishing different materials [28], [29]. To combine the advantages of two data sources and remove the complex interaction of illumination and surface geometry associated with spectral information [41], [42], it is necessary to consider the intrinsic relationship between HSI and LiDAR elevation information.

Intrinsic information recovery can effectively eliminate the spectral variation caused by illumination changes; however, it assumes a uniform light and neglects shadows in the entire scene. Objects in shaded regions usually have much lower intensity than that in illuminated regions and, therefore, would cause difficulty in recognition and classification. In this article, we propose a novel IID model for hyperspectral point cloud that can detect the shaded regions and recover their corresponding reflectance. We first estimate the global illumination of the scene using an intrinsic information recovery method. Then, we perform supervoxel segmentation on hyperspectral point cloud, calculate the blocking relation of supervoxels, and therefore accurately detect shaded regions. Finally, we estimate the illumination and reflectance of the shaded region based on an illumination-invariant spectral prior. The experimental results show that the proposed method can effectively detect shaded areas and generate shadow-less hyperspectral point cloud.

In summary, our main contribution is twofold.

1) *Shadow Region Detection by Calculating Blocking Relations:* By projecting each supervoxel along the light directions, we calculate the blocking relations of different objects. Supervoxels that are blocked are recognized as objects lying in a shaded area. The advantage of our approach is that it would not confuse those dark objects with blocked ones.
2) *Illumination-Invariant Reflectance Prior:* We introduce an illumination-invariant reflectance prior that characterizes the similarity between objects in lit and shaded regions. In this way, we ensure the reflectance consistency globally and improve the shadow removal performance.

The rest of this article is introduced as follows. In Section II, we introduce some related work, and our model is introduced in great detail in Section III. Section IV presents the experimental results and related analysis. Conclusions of this article are given in Section V.

II. Related Work

Combining the advantages of LiDAR and HSI data sources is to acquire abundant spectral and geometric information simultaneously, and researchers have proposed a variety of HSI and LiDAR fusion methods. There are three main categories that are based on pixels, features, or decision levels. In more detail, as for pixel-level methods, point clouds are converted to digital surface model (DSM) with elevation information, which is further appended to the spectral dimension of HSI as a new image channel [30], for example, Ghamisi and Hoefle [31] converted LiDAR point clouds to 2-D information, such as laser echo ratio, variance of point elevation, plane fitting residuals, and echo intensity; based on pixel-level methods, feature-level methods usually merge geometric features of DSM and spectral features of HSI [32], [33], [34], [35]. In [33], each point is represented as a second-order tensor, which is projected into the feature space to extract the discriminative geometric-spectral features; the decision-level methods perform the classification tasks on LiDAR and HSI and finally merge their classification labels together [37], and the combined data can be interpreted in [36], by coloring and texturing the lidar geometry with hyperspectral mineral maps.

Generally, compared with the most current HSI and LiDAR fusion methods, there are some connections and differences between our work and data fusion methods. As for connection part, both are motivated by the consideration of making full use of these two complementary data and taking advantage of every data source. As for the difference part, it can be grouped together on three levels. First, fusion methods convert and encapsulate 3-D point cloud data into 2-D images [38], [39], which will help to calculate and analyze different data, but will also lead to loss of information [40]. While the unorganized point cloud data are directly used in this article, the relationship between HSI data and LiDAR point clouds data is explored to give full play to complementarity between two data sources. Second, the existing fusion methods mainly pay more attention on heterogeneity than consistency between HSI and LiDAR data, and since reflectance only corresponds to the material properties, it is necessary to do an IID operation to recover material reflectance from the complex environmental interaction result, which is considered in our work. Third, fusion methods tend to extract human-designed features from two datasets and then consider how to combine them to offer valuable information, while in this article, we physically infer the properties that are directly related to HSI and LiDAR point cloud data.

Many shadow removal studies have been done to detect shadow areas and remove them from single optical image [45]

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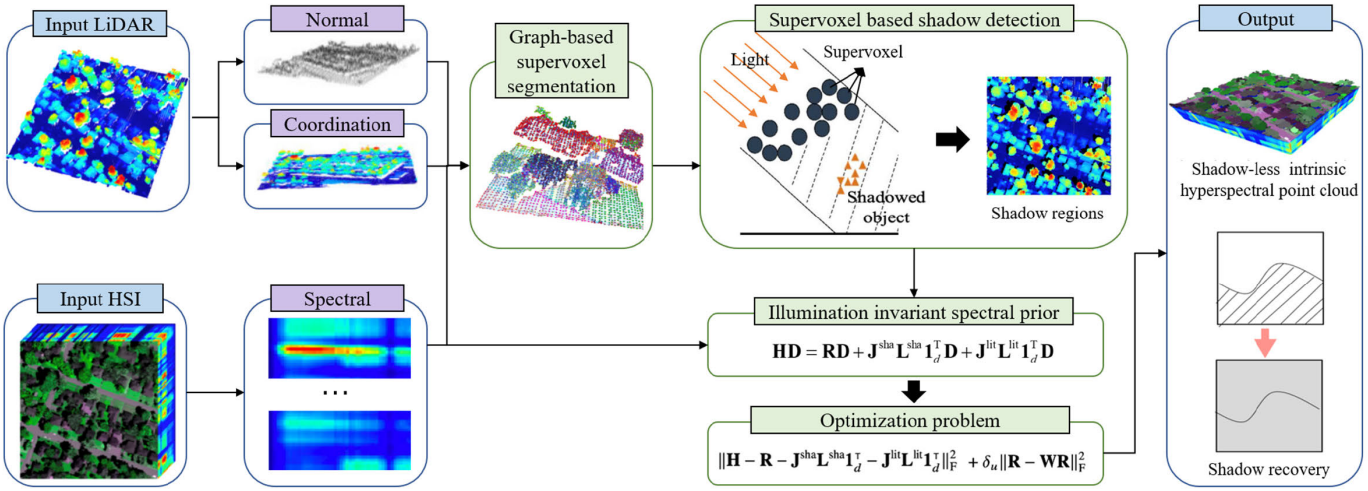


Fig. 1. Flowchart of the proposed method.

and first developed and easiest to handle with shadow removal problem focused on RGB images [46], [47], [48], [49]. For instance, Arbel and Hel-Or modeled an illumination recovery problem based on the idea of 3-D surface reconstruction to remove edges between shadow and nonshadow area intensities and successfully remove shadow of a single RGB image [47]. With the development of hyperspectral imaging techniques, some shadow removal methods for hyperspectral images (HSIs) have been studied continuously. As the number of spectral bands increases dramatically, it is more difficult to process HSIs than RGB. Duan et al. [50] designed a shadow removal framework for the characteristics of hyperspectral data, in which the principle of intrinsic decomposition model is exploited to successfully estimate the spectral variations between shadow and nonshadow regions in different exposure levels. Hartzell et al. [51] designed a shadow restoration framework that fuses the HSI with terrestrial laser scanning point cloud data to detect shadowed HSI pixels via geometric ray tracing. However, these methods still have drawbacks in which image space shadow detection methods depend on threshold value or spectral features, which leads to weak robustness for different scenes; besides, existing methods using ray tracing occlusion analysis spend more time on the calculation of occlusion relation. Similar to fusion work, existing shadow removal works do not pursue reflectance recovery of HSIs.

To address the problems faced by these related approaches aforementioned, we provide a hyperspectral point cloud IID method, which aims at detecting shaded regions and recovering reflectance information from complex spectral information by fully mining the relation between HSI and LiDAR point cloud data.

III. Shadow-Less Intrinsic Hyperspectral Point Cloud Generation

In this section, we will describe our model in detail. Fig. 1 shows the overall flowchart of this model, which consists of several main steps. First, the supervoxel segmentation is applied to hyperspectral point cloud. Second, the shaded areas are detected based on the segmented data. Then, both shadow regions and illumination-invariant spectral prior are considered in the hyperspectral imaging model. Finally, IHSPC is generated with a built model.

A. Hyperspectral Point Cloud Supervoxel Segmentation

Given hyperspectral point cloud $\mathbf{P} = [\mathbf{p}_1, \mathbf{p}_2, \ldots, \mathbf{p}_v]^{\mathrm{T}} \in \mathbb{R}^{v\times(3+d)}$ and $\mathbf{p}_i = [x_i, y_i, z_i, \rho(\lambda_1), \rho(\lambda_2), \ldots, \rho(\lambda_d)]^{\mathrm{T}}$, where (x_i, y_i, z_i) is the 3d coordinate of each point and $(\rho(\lambda_1), \rho(\lambda_2), \ldots, \rho(\lambda_d))$ is the spectral information of each point, $S = \{S_i\}_{i=1}^{t}$ is the partition of supervoxel oversegmentation processed on $\mathbf{P}$ such that $\mathbf{P} = \bigcup_i^t S_i$, where $S_i \cap S_j = \varnothing$ when $i \neq j$ and t is the number of supervoxels.

First, we need to compute the normal of each point. Assume that a certain point in the point cloud, $\mathbf{p}_i$, lies on a local surface point set $\{\mathbf{p}_j\}_{j=1}^{k}$. We use $N^x x + N^y y + N^z z + D = 0$ to fit surface point set $\{\mathbf{p}_j\}_{j=1}^{k}$, where $\mathbf{N} = [N^x, N^y, N^z]^{\mathrm{T}}$ denotes the surface normal. The error of fitting is then calculated by

$$\mathrm{dis} = \sum_{j=1}^{k} \left(\mathbf{P}_j^{\mathrm{T}}\mathbf{N} + D\right)^2, \quad \mathbf{N}^{\mathrm{T}}\mathbf{N} = 1. \tag{2}$$

Minimize dis, and then, we have

$$\arg\min_{\mathbf{N}} \sum_{j=1}^{k} \left(\mathbf{P}_j^{\mathrm{T}}\mathbf{N} + D\right)^2, \quad \text{s.t. } \mathbf{N}^{\mathrm{T}}\mathbf{N} = 1 \tag{3}$$

$$\mathbf{N}^{\mathrm{T}}\left(\sum_{j=1}^{k}\mathbf{P}_j\mathbf{P}_j^{\mathrm{T}}\right)\mathbf{N} + 2D\sum_{j=1}^{k}\mathbf{P}_j^{\mathrm{T}}\mathbf{N} + kD^2 - \delta\left(\mathbf{N}^{\mathrm{T}}\mathbf{N} - 1\right) = 0. \tag{4}$$

Solving (4), we obtain the normal of the point cloud.

To get a segment result that is spectrally and spatially homogeneous, we introduce the idea of weighted graph clustering to solve the supervoxel segmentation problem. Initially, we set v as the number of hyperspectral point clouds, in which case S_i denotes the ith hyperspectral point cloud. Therefore, we let $G = (V, E)$ denote the weighted graph and initialize it by following the steps given next: first, we let node $v_i \in V$ denote

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a hyperspectral point cloud $\mathbf{p}_i$; second, if both nodes i and j satisfy the inequality $\|(x_i, y_i, z_i) - (x_j, y_j, z_j)\| < \delta_x$, they will be joined by an edge, where $\|(x_i, y_i, z_i) - (x_j, y_j, z_j)\|$ calculates the Euclidean distance between two hyperspectral point clouds and δ_x defines the radius of the neighboring region of each hyperspectral point; and third, the weight of nodes i and j joined by edge e is measured by

$$\begin{aligned} W(i,j) &= \alpha\left\|\left(\rho_i(\lambda_1),\ldots,\rho_i(\lambda_d)\right) - \left(\rho_j(\lambda_1),\ldots,\rho_j(\lambda_d)\right)\right\| \\ &\quad + \beta\left\|z_i - z_j\right\| + \gamma\left\|\left(N_i^x, N_i^y, N_i^z\right) - \left(N_j^x, N_j^y, N_j^z\right)\right\|. \end{aligned} \tag{5}$$

We further give the supplementary supervoxel clustering strategy that: fourth, all edges in E are simply sorted by weight from smallest to largest; finally, we check whether two nodes joined by edge e in the sorted list belong to different supervoxels and the inequality $W(e) < \mathrm{Dis}(S_i, S_j) + \delta_I$ is satisfied, if so, merge two supervoxels S_i and S_j, where $\mathrm{Dis}(S_i, S_j)$ is the smallest weight of S_i and S_j, and δ_I is a threshold.

B. Supervoxel-Based Blocking Shadow Detection

Since the light source is not irradiated vertically on the surface of the ground object, objects with a certain height, such as buildings and trees, block the light source, forming smaller shadows scattered throughout the image. Existing shadow detection methods mainly use hyperspectral spectral information to extract features such as chromaticity, texture, and edge to distinguish shadow and nonshadow regions. These methods work well for larger shadows but are not so good for shadows due to occlusion between objects. Since some small targets also have darker spectral values, these targets are often indistinguishable by feature and threshold methods.

For stereo spectral data, whether it is a hyperspectral point cloud or a hyperspectral voxel, it is very difficult to calculate the occlusion relationship between each point cloud or each voxel. Therefore, the method uses supervoxels as the basic unit to calculate the occlusion relationship between each voxel. More specifically, for each supervoxel, we project it onto the horizontal ground according to the light direction and then compute the envelope of each supervoxel projected on the horizontal ground. According to the inclusion or overlapping relationship between the envelopes and the height of the supervoxel corresponding to the envelope, it can be judged whether a supervoxel is occluded by another supervoxel, and the occluded supervoxel belongs to a shaded area.

We use an intrinsic information recovery method to estimate the light direction $\mathbf{L} \in \mathbb{R}^3$ of the HSIs. Since supervoxels are homogeneous spatially and spectrally, we assume that hyperspectral point clouds in each supervoxel should have the same reflectance value. Therefore, the supervoxel-based hyperspectral reflectance component is defined as $\hat{\mathbf{R}} = [\hat{R}_1, \hat{R}_2, \ldots, \hat{R}_t]^{\mathrm{T}} \in \mathbb{R}^{t\times d}$.

We let $\mathbf{h}_a$ and $\mathbf{h}_b$ represent an HSI pixel pair, and the following equation system will be satisfied if they fall completely in the same supervoxel [43]:

$$\begin{cases} \mathbf{L}^{\mathrm{T}}\left(\sum_{j=1}^{v}\hat{A}_{aj}\mathbf{N}_j\right)\hat{R}(\lambda_k) = h_a(\lambda_k) \\ \mathbf{L}^{\mathrm{T}}\left(\sum_{j=1}^{v}\hat{A}_{bj}\mathbf{N}_j\right)\hat{R}(\lambda_k) = h_b(\lambda_k) \end{cases} \quad k = 1, 2, \ldots, d \tag{6}$$

where $\hat{A}_{aj}$ and $\hat{A}_{bj}$ represent the light intensity on point cloud $\mathbf{p}_j$ distributed by HSI pixel $\mathbf{h}_a$ and $\mathbf{h}_b$, respectively. Furthermore, we have

$$\frac{1}{\hat{R}(\lambda_k)} = \mathbf{L}^{\mathrm{T}}\sum_{j=1}^{v}\frac{\hat{A}_{aj}}{h_a(\lambda_k)}\mathbf{N}_j = \mathbf{L}^{\mathrm{T}}\sum_{j=1}^{v}\frac{\hat{A}_{bj}}{h_b(\lambda_k)}\mathbf{N}_j. \tag{7}$$

We can see that $\hat{R}(\lambda_k)$ can be eliminated by the above equation and get an overdetermined equation about $\mathbf{L}$

$$\mathbf{L}^{\mathrm{T}}\sum_{j=1}^{v}\left(\left(\frac{\hat{A}_{aj}}{h_a(\lambda_k)} - \frac{\hat{A}_{bj}}{h_b(\lambda_k)}\right)\mathbf{N}_j\right) = 0. \tag{8}$$

Obviously, $\mathbf{L}$ can be estimated by solving the overdetermined equation system given by (8).

For supervoxel $S_1(x_i, y_i, z_i)$, $i = 1, 2, \ldots, k$, let the projection of S_1 along the light direction on the horizontal ground be $S_1'(x_i, y_i)$, $i = 1, 2, \ldots, k$. Then, we assume that the elliptic envelope of S_1' is a smooth curve $F_1(x, y, c) = 0$. To facilitate the solution, we further assume that $F_1(x, y, c) = 0$ is an ellipse

$$\begin{aligned} F_1(x, y, a_1, b_1, A_1, B_1, C_1) &= \left(\frac{x-a_1}{A_1}\right)^2 + \left(\frac{y-b_1}{B_1}\right)^2 - C_1^2 \\ &= 0. \end{aligned} \tag{9}$$

Therefore, $F_1(x, y, a_1, b_1, A_1, B_1, C_1)$ is equivalent to solving the following problem:

$$\begin{aligned} &\arg\min\ F_1(x, y, a_1, b_1, A_1, B_1, C_1) \\ &\text{s.t. } F_1(x_i, y_i, a_1, b_1, A_1, B_1, C_1) < 0, \quad i = 1, 2, \ldots, k. \end{aligned} \tag{10}$$

Given two known supervoxels, S_1 and S_2, and their ellipse envelopes projected on the horizontal ground, $\mathrm{ellipse}(a_1, b_1, A_1, B_1, C_1)$ and $\mathrm{ellipse}(a_2, b_2, A_2, B_2, C_2)$, we can calculate whether there is an occlusion relationship between them. Supervoxel S_2 is occluded by S_1 if and only if

$$\begin{cases} F_1(a_2 + A_2C_2, b_2 + B_2C_2, a_1, b_1, A_1, B_1, C_1) < 0 \\ F_1(a_2 + A_2C_2, b_2 - B_2C_2, a_1, b_1, A_1, B_1, C_1) < 0 \\ F_1(a_2 - A_2C_2, b_2 + B_2C_2, a_1, b_1, A_1, B_1, C_1) < 0 \\ F_1(a_2 - A_2C_2, b_2 - B_2C_2, a_1, b_1, A_1, B_1, C_1) < 0 \\ \left(\bar{z}\right)_2 - \left(\bar{z}\right)_1 < 0 \end{cases} \tag{11}$$

where $(\bar{z})_1$ and $(\bar{z})_2$ are the average heights of supervoxels S_1 and S_2, respectively. If the supervoxel S_2 is occluded by S_1, then the supervoxel S_2 is in the occluded shaded area.

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C. Hyperspectral Imaging Model Considering Shadows

First, we use α_j to represent the distribution of shadows

$$\begin{cases} \alpha_j = 1, & \text{when } \mathbf{p}_j \text{ is in shadow} \\ \alpha_j = 0, & \text{otherwise} \end{cases} \tag{12}$$

and when $\alpha_j = 1$, it means that the point $\mathbf{p}_j$ belongs to the shaded area, and when $\alpha_j = 0$, it means that the point $\mathbf{p}_j$ does not belong to the shaded area. Let $\mathbf{L}^{\text{sha}}$ be the illumination of the shaded area, and let $\mathbf{J}^{\text{sha}} = [\alpha_1\mathbf{J}_1, \alpha_2\mathbf{J}_2, \ldots, \alpha_v\mathbf{J}_v]^{\mathrm{T}} \in \mathbb{R}^{v\times 9}$ be the geometric component of the shaded area; $\mathbf{L}^{\text{sha1}} = [L_1^{\text{sha1}}, L_2^{\text{sha1}}, \ldots, L_9^{\text{sha1}}]^{\mathrm{T}}$ is the first- and second-order spherical harmonic illumination coefficients and $\mathbf{J}_j = [c_4, 2c_2N_j^y, 2c_2N_j^x, c_2N_j^z, 2c_1N_j^xN_j^y, 2c_1N_j^yN_j^z, c_3N_j^zN_j^z - c_5, 2c_1N_j^xN_j^z, c_1N_j^xN_j^x - c_1N_j^yN_j^y]^{\mathrm{T}}$, where N_j is the normal of point j.

Similarly, we assume that the illumination of the nonshaded area is $\mathbf{L}^{\text{lit}}$, and the geometric component is $\mathbf{J}^{\text{lit}} = [(1-\alpha_1)\mathbf{J}_1, (1-\alpha_2)\mathbf{J}_2, \ldots, (1-\alpha_v)\mathbf{J}_v]^{\mathrm{T}} \in \mathbb{R}^{v\times 9}$. Therefore, in the case of considering shadows, the point cloud-level hyperspectral imaging model is formulated as follows:

$$\mathbf{H} = \mathbf{R} + \mathbf{J}^{\text{sha}}\mathbf{L}^{\text{sha}}\mathbf{1}_d^{\mathrm{T}} + \mathbf{J}^{\text{lit}}\mathbf{L}^{\text{lit}}\mathbf{1}_d^{\mathrm{T}} \tag{13}$$

where $\mathbf{J}^{\text{sha}}\mathbf{L}^{\text{sha}}\mathbf{1}_d^{\mathrm{T}}$ is the shadow term, $\mathbf{J}^{\text{lit}}\mathbf{L}^{\text{lit}}\mathbf{1}_d^{\mathrm{T}}$ is the non-shaded term, and $\mathbf{1}_d$ denotes the $d \times 1$ vector with all 1s.

D. Illumination-Invariant Spectral Prior

This further describes their spectral similarity for the same material in shaded and nonshaded regions. Examining the hyperspectral imaging (13) under shadow conditions, in order to seek a certain illumination-invariant characteristic, we can try to find a mapping matrix $\mathbf{D}$ to remove the influence of illumination

$$\begin{aligned} \mathbf{HD} &= \mathbf{RD} + \mathbf{J}^{\text{sha}}\mathbf{L}^{\text{sha}}\mathbf{1}_d^{\mathrm{T}}\mathbf{D} + \mathbf{J}^{\text{lit}}\mathbf{L}^{\text{lit}}\mathbf{1}_d^{\mathrm{T}}\mathbf{D} \\ &= \mathbf{RD}. \end{aligned} \tag{14}$$

This is to find the mapping matrix $\mathbf{D}$ such that

$$\left(\mathbf{J}^{\text{sha}}\mathbf{L}^{\text{sha}} + \mathbf{J}^{\text{lit}}\mathbf{L}^{\text{lit}}\right)\mathbf{1}_d^{\mathrm{T}}\mathbf{D} = \mathbf{0}. \tag{15}$$

Suppose further that $\mathbf{D}$ is an idempotent symmetric matrix. According to the characteristics of an idempotent symmetric matrix, the eigenvalue of $\mathbf{D}$ can only be 0 or 1, and the trace of $\mathbf{D}$ is equal to the rank of $\mathbf{D}$. According to this characteristic, we can construct the matrix $\mathbf{D}$ as follows:

$$\mathbf{D} = \mathbf{I} - \mathbf{X}^{-1}\begin{bmatrix} 1 & & & \\ & 0 & & \\ & & \ddots & \\ & & & 0 \end{bmatrix}\mathbf{X}$$

$$\mathbf{X} = \begin{bmatrix} x_1 & x_2 & \cdots & x_d \\ -x_2 & x_1 & & \\ \vdots & & \ddots & \\ -x_d & & & x_1 \end{bmatrix}. \tag{16}$$

Then, we have

$$\mathbf{D} = \mathbf{I} - \frac{\mathbf{x}\mathbf{x}^{\mathrm{T}}}{\mathbf{x}^{\mathrm{T}}\mathbf{x}} \tag{17}$$

where $\mathbf{x} = [x_1, x_2, \ldots, x_d]^{\mathrm{T}}$ is the coefficient to be found.

In order to find $\mathbf{x} = [x_1, x_2, \ldots, x_d]^{\mathrm{T}}$, we need to find a set of points $\{\mathbf{H}_i\}_{i=1}^k$ with the same reflectivity distributed in the shaded area and the nonshaded area.

By (14)

$$\mathbf{H}_1\left(\mathbf{I} - \frac{\mathbf{x}\mathbf{x}^{\mathrm{T}}}{\mathbf{x}^{\mathrm{T}}\mathbf{x}}\right) = \mathbf{H}_2\left(\mathbf{I} - \frac{\mathbf{x}\mathbf{x}^{\mathrm{T}}}{\mathbf{x}^{\mathrm{T}}\mathbf{x}}\right) = \cdots = \mathbf{H}_k\left(\mathbf{I} - \frac{\mathbf{x}\mathbf{x}^{\mathrm{T}}}{\mathbf{x}^{\mathrm{T}}\mathbf{x}}\right) \tag{18}$$

and furthermore,

$$\bar{\mathbf{H}}_1\left(\mathbf{I} - \frac{\mathbf{x}\mathbf{x}^{\mathrm{T}}}{\mathbf{x}^{\mathrm{T}}\mathbf{x}}\right) = \bar{\mathbf{H}}_2\left(\mathbf{I} - \frac{\mathbf{x}\mathbf{x}^{\mathrm{T}}}{\mathbf{x}^{\mathrm{T}}\mathbf{x}}\right) = \cdots = \bar{\mathbf{H}}_k\left(\mathbf{I} - \frac{\mathbf{x}\mathbf{x}^{\mathrm{T}}}{\mathbf{x}^{\mathrm{T}}\mathbf{x}}\right) = 0 \tag{19}$$

where $\bar{\mathbf{H}}_i = \mathbf{H}_i - (\sum_{j=1}^k \mathbf{H}_j)/k$.

Solving the system of (19) is equivalent to minimizing the following loss function:

$$\begin{aligned} F &= \sum_{i=1}^k \bar{\mathbf{H}}_i^{\mathrm{T}}\mathbf{D}^2\bar{\mathbf{H}}_i \\ &= \sum_{i=1}^k \bar{\mathbf{H}}_i^{\mathrm{T}}\mathbf{D}\bar{\mathbf{H}}_i \\ &= \sum_{i=1}^k \bar{\mathbf{H}}_i^{\mathrm{T}}\bar{\mathbf{H}}_i - \frac{\mathbf{x}^{\mathrm{T}}\left(\sum_{i=1}^k \bar{\mathbf{H}}_i\bar{\mathbf{H}}_i^{\mathrm{T}}\right)\mathbf{x}}{\mathbf{x}^{\mathrm{T}}\mathbf{x}}. \end{aligned} \tag{20}$$

Minimizing F is equivalent to maximizing the Rayleigh quotient $\text{Rq}(\mathbf{M}, \mathbf{x}) = ((\mathbf{x}^{\mathrm{T}}(\sum_{i=1}^k \bar{\mathbf{H}}_i\bar{\mathbf{H}}_i^{\mathrm{T}})\mathbf{x})/(\mathbf{x}^{\mathrm{T}}\mathbf{x}))$, where $\mathbf{M} = (\sum_{i=1}^k \bar{\mathbf{H}}_i\bar{\mathbf{H}}_i^{\mathrm{T}})$. According to the properties of Rayleigh entropy, the maximum value of $\text{Rq}(\mathbf{M}, \mathbf{x})$ is equal to the largest eigenvalue of $\mathbf{M}$, and the minimum value of $\text{Rq}(\mathbf{M}, \mathbf{x})$ is equal to the smallest eigenvalue of $\mathbf{M}$, i.e.,

$$\lambda_{\min} \le \text{Rq}(\mathbf{M}, \mathbf{x}) \le \lambda_{\max}. \tag{21}$$

Therefore, $\mathbf{x}$ required in this article is the eigenvector that corresponds to the largest eigenvalue of $\mathbf{M} = (\sum_{i=1}^k \bar{\mathbf{H}}_i\bar{\mathbf{H}}_i^{\mathrm{T}})$.

By encoding the HSI $\mathbf{HD}$ mapped with the matrix $\mathbf{D}$ by ℓ^1 graph coding [19], we can learn the spectral prior that the illumination is invariant

$$\mathbf{R} = \mathbf{WR}. \tag{22}$$

E. Recovery of Reflectance

In the case of considering shaded area, before recovering the reflectance and ambient light in the shaded area, an optimization problem requires to be solved

$$\begin{aligned} \arg\min_{\mathbf{R},\mathbf{L}^{\text{sha}},\mathbf{L}^{\text{lit}}} U = &\left\|\mathbf{H} - \mathbf{R} - \mathbf{J}^{\text{sha}}\mathbf{L}^{\text{sha}}\mathbf{1}_d^{\mathrm{T}} - \mathbf{J}^{\text{lit}}\mathbf{L}^{\text{lit}}\mathbf{1}_d^{\mathrm{T}}\right\|_F^2 \\ &+ \delta_\mu\|\mathbf{R} - \mathbf{WR}\|_F^2. \end{aligned} \tag{23}$$

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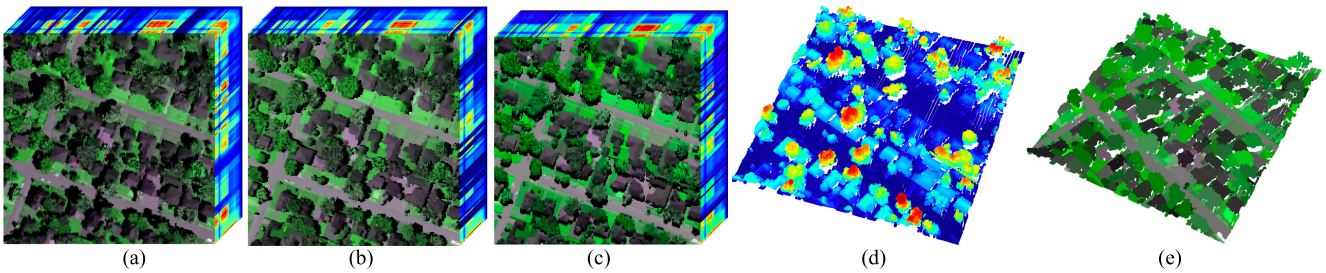


Fig. 2. Pseudo-synthetic HSIs and LiDAR with ground-truth hyperspectral point cloud. (a) HSI_light1, (b) HSI_light2, and (c) HSI_light are the three synthetic HSIs of different light conditions. (d) Height map of LiDAR point cloud. (e) Ground-truth IHSPC.

Finding the first-order partial derivative for $U(\mathbf{R}, \mathbf{L}^{\text{sha}}, \mathbf{L}^{\text{lit}})$, we have

$$\begin{cases} \dfrac{1}{2}\dfrac{\partial U}{\partial \mathbf{R}} = \mathbf{G}\mathbf{R} + \mathbf{J}^{\text{sha}}\mathbf{L}^{\text{sha}}\mathbf{1}_d^{\text{T}} + \mathbf{J}^{\text{lit}}\mathbf{L}^{\text{lit}}\mathbf{1}_d^{\text{T}} - \mathbf{H} \\ \mathbf{G} = \mathbf{I}_v + \delta_\mu\left(\mathbf{I}_v - \mathbf{W}^{\text{T}}\right)\left(\mathbf{I}_v - \mathbf{W}\right) \\ \dfrac{1}{2}\dfrac{\partial U}{\partial \mathbf{L}^{\text{sha}}} = d\left(\mathbf{J}^{\text{sha}}\right)^{\text{T}}\mathbf{J}^{\text{sha}}\mathbf{L}^{\text{sha}} \\ \qquad\qquad + d\left(\mathbf{J}^{\text{sha}}\right)^{\text{T}}\mathbf{J}^{\text{lit}}\mathbf{L}^{\text{lit}} - \left(\mathbf{J}^{\text{sha}}\right)^{\text{T}}(\mathbf{H} - \mathbf{R})\mathbf{1}_d \\ \dfrac{1}{2}\dfrac{\partial U}{\partial \mathbf{L}^{\text{lit}}} = d\left(\mathbf{J}^{\text{lit}}\right)^{\text{T}}\mathbf{J}^{\text{lit}}\mathbf{L}^{\text{lit}} \\ \qquad\qquad + d\left(\mathbf{J}^{\text{lit}}\right)^{\text{T}}\mathbf{J}^{\text{sha}}\mathbf{L}^{\text{sha}} - \left(\mathbf{J}^{\text{lit}}\right)^{\text{T}}(\mathbf{H} - \mathbf{R})\mathbf{1}_d \end{cases} \tag{24}$$

where $\mathbf{I}_v$ denotes the $v \times v$ identity matrix. Let $\partial U/\partial \mathbf{R} = 0$, $\partial U/\partial \mathbf{L}^{\text{sha}} = 0$, and $\partial U/\partial \mathbf{L}^{\text{lit}} = 0$, and the overdetermined equation system can be obtained

$$\begin{bmatrix} \mathbf{G} & \mathbf{J}^{\text{sha}} & \mathbf{J}^{\text{lit}} \\ \dfrac{(\mathbf{J}^{\text{sha}})^{\text{T}}}{d} & (\mathbf{J}^{\text{sha}})^{\text{T}}(\mathbf{J}^{\text{sha}}) & (\mathbf{J}^{\text{sha}})^{\text{T}}\mathbf{J}^{\text{lit}} \\ \dfrac{(\mathbf{J}^{\text{lit}})^{\text{T}}}{d} & (\mathbf{J}^{\text{lit}})^{\text{T}}(\mathbf{J}^{\text{lit}}) & (\mathbf{J}^{\text{lit}})^{\text{T}}\mathbf{J}^{\text{sha}} \end{bmatrix} \begin{bmatrix} \bar{\mathbf{R}} \\ \mathbf{L}^{\text{sha}} \\ \mathbf{L}^{\text{lit}} \end{bmatrix} = \begin{bmatrix} \bar{\mathbf{H}} \\ \dfrac{(\mathbf{L}^{\text{sha}})^{\text{T}}\mathbf{H}}{d} \\ \dfrac{(\mathbf{L}^{\text{lit}})^{\text{T}}\mathbf{H}}{d} \end{bmatrix} \tag{25}$$

where $\bar{\mathbf{H}} = (1/d)\mathbf{H}\mathbf{1}_d$ and $\bar{\mathbf{R}} = (1/d)\mathbf{R}\mathbf{1}_d$ are the arithmetic mean of $\mathbf{H}$ and $\mathbf{R}$, respectively.

By solving (25), we can quickly get the ambient light of the shaded area $\mathbf{L}^{\text{sha}}$, nonshaded area $\mathbf{L}^{\text{lit}}$, and shadow-less IHSPC reflectance $\mathbf{R}$.

IV. Experimental Results and Analysis

In this section, experimental results on both pseudo-synthetic and real data are presented to assess the effectivity of this shadow-less intrinsic information recovery method for HSI and LiDAR.

A. Experiments on Pseudo-Synthetic Data

1) *Data Description:* Obviously, it is difficult to obtain the truth values of normals, illumination, and hyperspectral reflectance from real natural scenes. In order to quantitatively assess the capability of this method, we produce several sets of pseudo-synthetic subscenes by changing illumination condition, which is shown in first three subfigures of Fig. 2, and last two subfigures show LiDAR data and corrsponding ground truth of IHSPC. The synthetic subscene is cropped from "grss_dfc_2018" dataset provided by the 2018 IEEE GRSS Data Fusion Contest, which is shown in Fig. 3 [44]. We generate intrinsic hyperspectral point cloud data by three steps: 1) a HSI image decomposition algorithm [19] is carried out on HSI to obtain a 2D reflectance image, 2) since the reflectance recovered by step one is still affected by the problem of spectral aliasing and shadows, we manually segment point clouds into different objects, 3) in the last step we manually map the spectrum of reflectance image to each point cloud object note that reflectance data from image boundaries and shaded areas should be ignored. After all the above processing steps, the generated reflectance component of hyperspectral point cloud can be treated as truth, and with the three-dimensional terrain, three-dimensional surface reflectance, a series of HSIs with different illuminations can be synthesized using discrete Monte Carlo method.

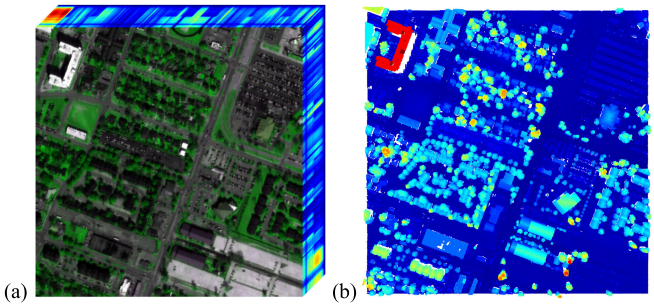


Fig. 3. Scene 272056_3289689 of "grss_dfc_2018" dataset. (a) HSI. (b) LiDAR point cloud.

2) *Results and Analysis:* Fig. 4 shows the supervoxel segmentation results obtained by the supervoxel segmentation method on the simulated data. We can see that the supervoxels without regularization obtained by the method in this article better maintain the boundaries of the image, but the shape of each supervoxel is not uniform, while the regularized supervoxels have uniform shape and scale. Fig. 5 shows the shaded areas detected by the method in this article on three sets of simulation data with different illuminations. As we can see,

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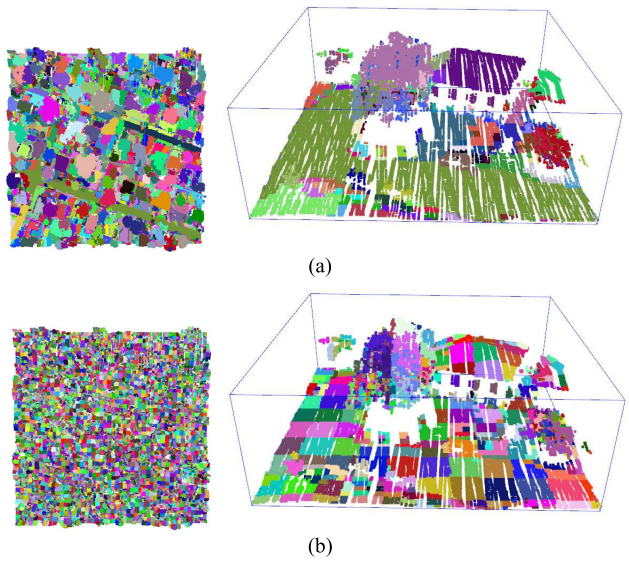


Fig. 4. Supervoxel segmentation results on synthetic data. (a) Not regularized. (b) Regularized.

shadows formed by light occlusion between buildings, trees, and other objects are detected accurately.

The shadow removal results obtained by our proposed method in detail are presented in Figs. 6–8. As we can see, Figs. 6–8 show the shadow recovery results of the proposed method on three simulated data with different light.

In fact, all three simulated hyperspectral datasets have obvious shadows formed by light occlusion between objects, especially the first simulated hyperspectral data with light1, which has the largest range of shadows due to the large inclination angle between light direction and the ground normal. Figs. 6(a), 7(a), and 8(a) show the IHSPC data with shadow generated using aforementioned steps; besides, Figs. 6(b), 7(b), and 8(b) show the recovery results in shaded regions obtained by the proposed method, from which we can see that the proposed method accurately detects the regions of shadows of three datasets with different illumination and effectively recovers the reflectance of shaded regions. Fig. 9 shows the root mean square error (RMSE) errors' comparison result of before and after recovering the reflectance information in shaded regions with the proposed method. Compared with simulated shaded data, the recovered hyperspectral point cloud data generated by the proposed method reduce the RMSE error by 20%–30%; moreover, it has more robust performance on datasets with different light and shadow conditions.

In this work, two state-of-the-art shadow removal methods are used for three synthetic data; they are shadow removal of HSIs with multiexposure fusion (HMF) [50] and shadow removal method using color lines (CLs) [52]. Parameters used in both methods are set according to authors' suggestions. An adaptive threshold method is used to generate an initial shadow detection map in the HMF method, and a linear shadow image model is used to estimate the color and illumination of the HSI for each channel in the CL method. We will analyze the comparison results visually in the following.

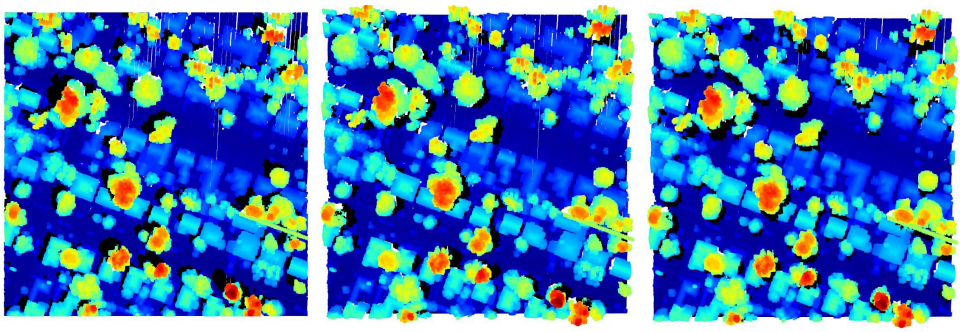
HMF can obtain shadow mask easily by threshold setting; however, as we can see from Fig. 10(b) that since the shadow area is so dark that it loses much spectral information, it is hard for HMF to recover the real spectrum after removing detected shadows. It can be seen from the shadow removal result of HMF that it requires artificial adjustment of threshold value, which is not suitable for all scenes; more importantly, it is difficult to recover the original spectrum when the shadow region is too dark and the spectral information is not enough. CL needs interaction to identify the shadow area, and although this is convenient, it is not suitable for scenes with large amount and small area of shadows. Similarly, as we can see from Fig. 10(c) that the shadow removal of CL is faced with the same problem as that of HMF method, shaded areas cannot be recovered completely. From Fig. 10(d), we can see that the shadow removal performance operated by the proposed method on three datasets is very obvious and successful since we detect shadow regions with supervoxel segmentation map, which is obtained based on spectral and 3-D spatial information extracted from HSI and LiDAR, respectively; in this way, sufficient physical occlusion information can be derived.

B. Experiments on Real Data

1) Data Description: It is a fact that it is extremely difficult to obtain real hyperspectral reflectance point cloud data, while another effective way to evaluate the shadow recovery result of hyperspectral point cloud data is to compare the 3-D object classification performance of recovered hyperspectral point cloud data generated by different methods, because intrinsic hyperspectral reflectance can improve the recognition ability of ground objects since various influence factors of imaging process, such as illumination condition, are removed. The data we use in this section are 272056_3289689 data, which is provided by the 2018 IEEE GRSS Data Fusion Contest. For 3-D classification, we select 0.5% of labeled samples for training and others for testing. Considering the variety of input features, the support vector machine is selected as our classifier, which is widely used and has strong robustness in classification performance. The different comparison methods are described as follows.

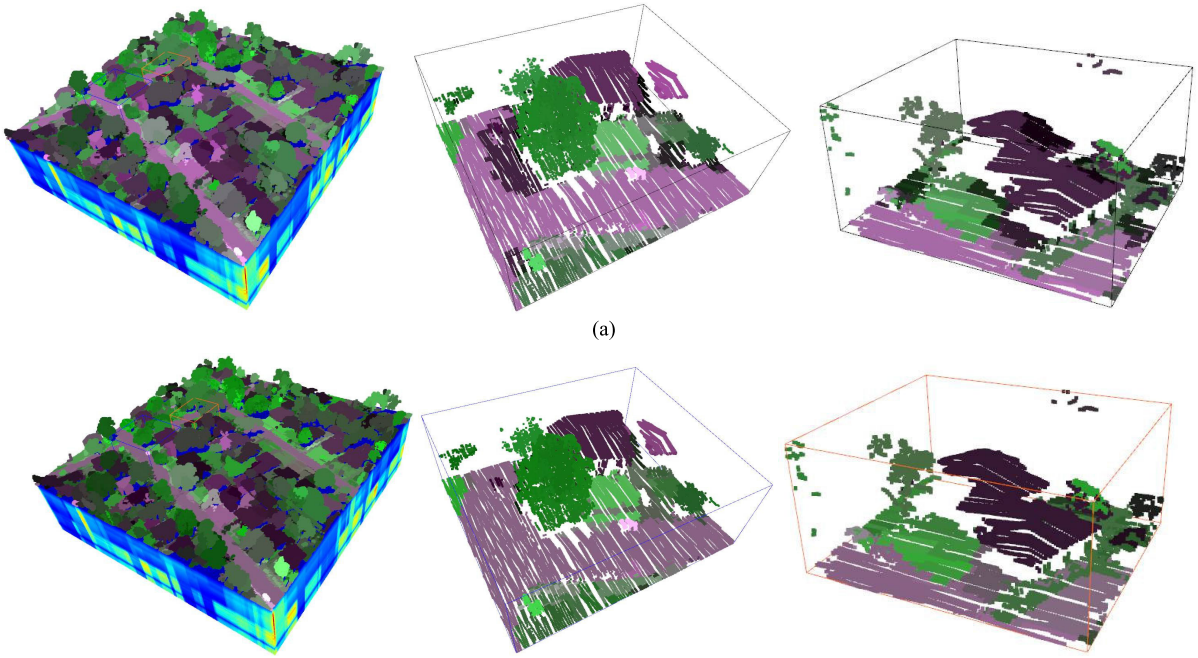
1) *MSPC:* Multispectral LiDAR point cloud (MSPC) data, each point has three feature channels of intensity.
2) *HSPC-Ori:* The hyperspectral point cloud data are generated by mapping the spectrum of HSI to LiDAR point through a lookup table, and each point has 48 spectral features.
3) *HSPC-2DRef:* The intrinsic decomposition method in [18] is used to recover the reflectance of HSI and then map its spectrum to the LiDAR point through a lookup table to generate IHSPC data; similarly, each point has 48 spectral features.
4) *Proposed:* The IHSPC data generated by our method, and each point has 48 spectral features.
5) *MSPC + Z:* Three spectral features generated by MSPC plus height features of point cloud data.
6) *HSPC-Ori + Z:* The 48 spectral features generated by HSPC-ori plus height features of point cloud data.

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(a) (b) (c)

Fig. 5. Shaded areas detected on synthetic data. (a) Light1. (b) Light2. (c) Light3.



(a)

(b)

Fig. 6. Shaded area recovery result of synthetic data-light1. (a) Hyperspectral point cloud with shadow. (b) Shaded area recovery result.

7) *HSPC-2DRef + Z:* The 48 spectral features generated by HSPC-2DRef plus height features of point cloud data.
8) *Proposed + Z:* The 48 spectral features generated by the proposed method plus height features of point cloud data.

2) Results and Analysis: Fig. 11 shows the result of supervoxel segmentation and shadow detection on real dataset, and Fig. 12 shows the rendering images of colored hyperspectral point cloud generated by MSPC, HSPC-ori, HSPC-2DRef, and our proposed method. In this article, we can see that the LiDAR point cloud generated by a multispectral LiDAR sensor does not suffer from the problem of illumination changes and shaded regions due to its active imaging system; moreover, there is no need to consider the registration problem between 3-D spatial information and spectral information since they are obtained simultaneously. However, MSPC has large noise due to the influence of laser scanning angle, also, there is an obvious specular reflection phenomenon occurring on smooth surface. In addition, as there are a small number of spectral channels of MSPC, it is difficult to expand it into hyperspectral point cloud due to the hardware limitation, while HSPC-ori is much more affected by factors such as illumination, shadow, and registration errors between HSIs and LiDAR point cloud. As for HSPC-2DRef, the influence of illumination factor can be reduced to some degree, but it still fails to remove shaded regions from the interested scene. In contrast, IHSPC generated by the proposed method effectively removes the influence of illumination changing and shaded regions simultaneously.

To illustrate the improved classification performance of this method in shaded regions, Fig. 13 shows the confusion matrix for the shaded samples. (The shaded regions are determined by selecting the darker part by thresholding.) From Fig. 13(a), we can see that the MSPC + Z method is not affected by shadow because of its active multispectral LiDAR imaging system, so its classification performance in the shaded area is similar to the overall classification performance in the entire scene; however, it completes the classification task

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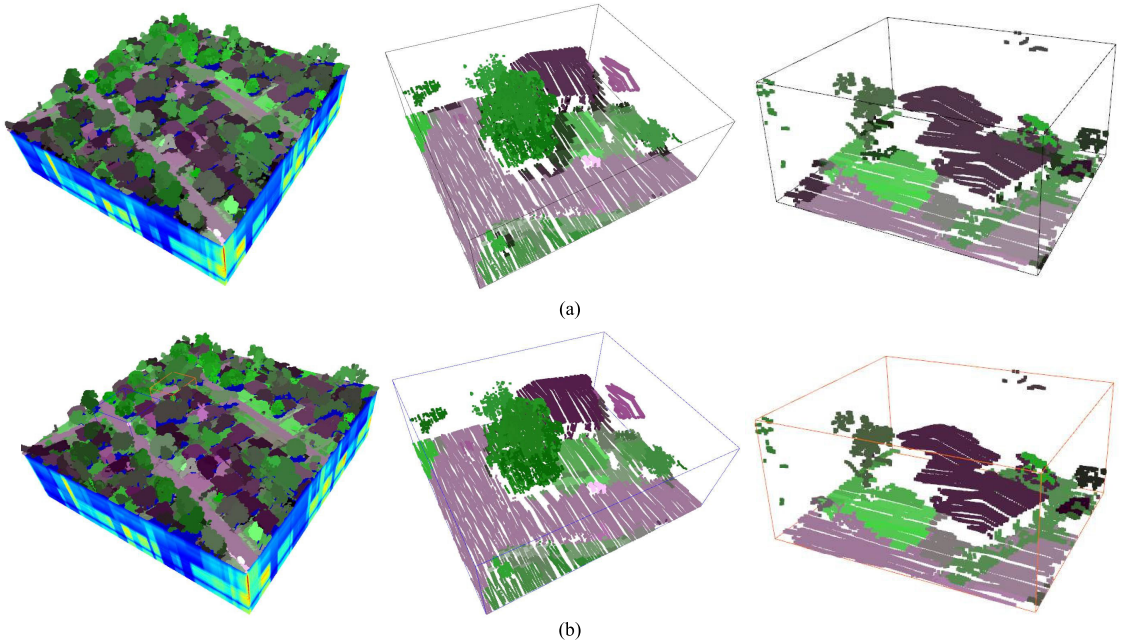


Fig. 7. Shadow recovery result of synthetic data-light2. (a) Hyperspectral point cloud with shadow. (b) Shadow recovery result.

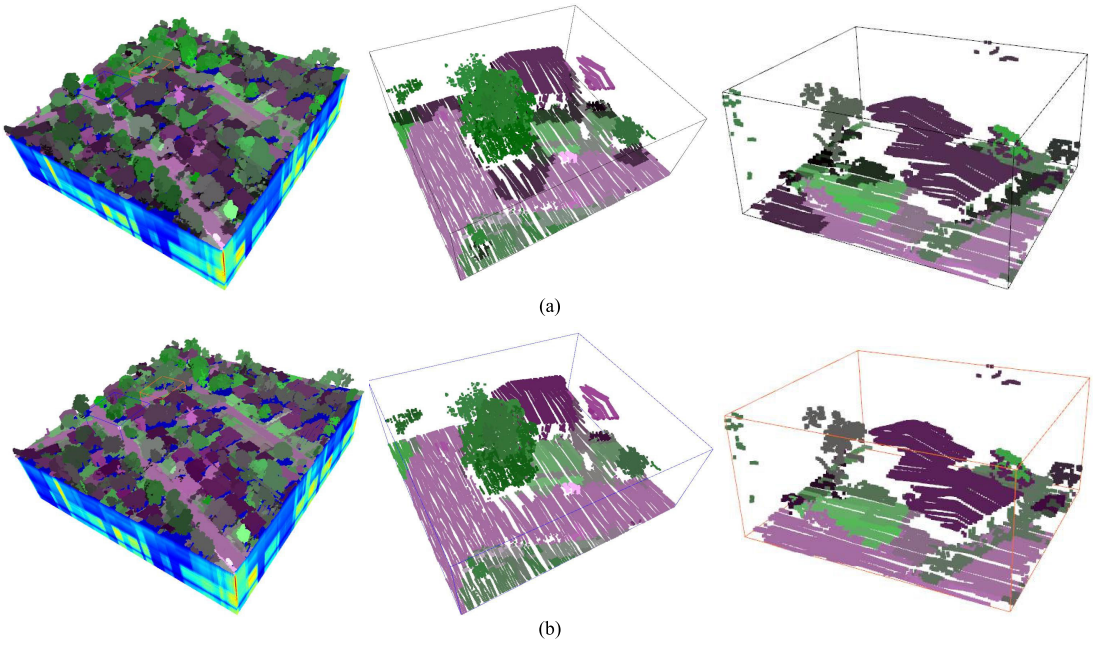


Fig. 8. Shadow recovery result of synthetic data-light3. (a) Hyperspectral point cloud with shadow. (b) Shadow recovery result.

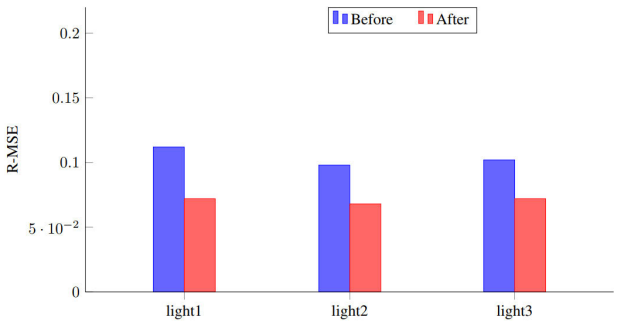


Fig. 9. RMSE errors of different methods on three synthetic data.

poorly, especially for tree class due to the attenuation of laser light scattering, and because there are only three channels of spectral bands in MSPC + Z, the recognition ability of ground objects is still lower than that of hyperspectral data. By analyzing Fig. 13(b) and (c), it is known that the classification accuracy of HSPC-ori + Z and HSPC-2DRef + Z methods for shaded regions is lower than that of the entire scene because they do not consider the effect of shadows on the spectrum information. Specifically, due to the influence of shadows, these two methods, HSPC-ori + Z and HSPC-2DRef + Z, have poor classification results for grass, roads, parking spaces, bare soil, impervious ground, and vehicles with less obvious height characteristics, which results in classification confusion between these categories. Fig. 13(d) shows that the reflectance of shaded regions can be recovered effectively with the proposed method so that the samples in

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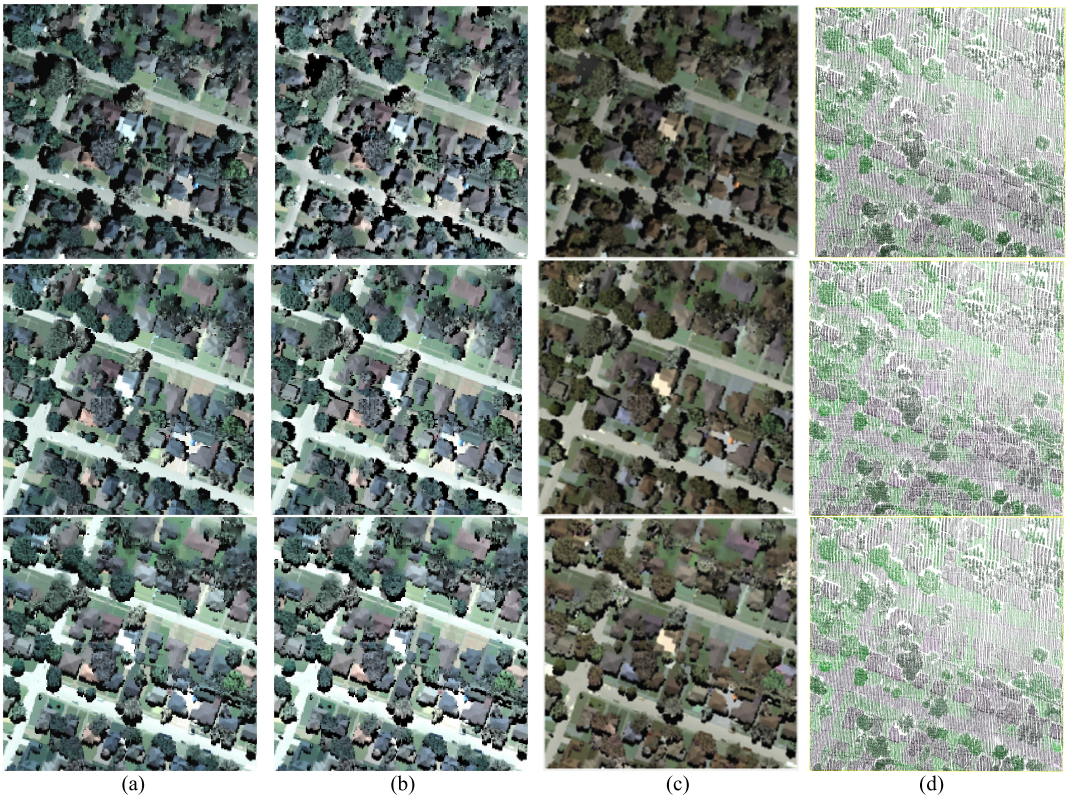


Fig. 10. Shadow removal comparison results on three synthetic datasets. From first row to bottom are data light1 to data light3. (a) Shaded scene. (b) HMF. (c) CL. (d) Proposed method.

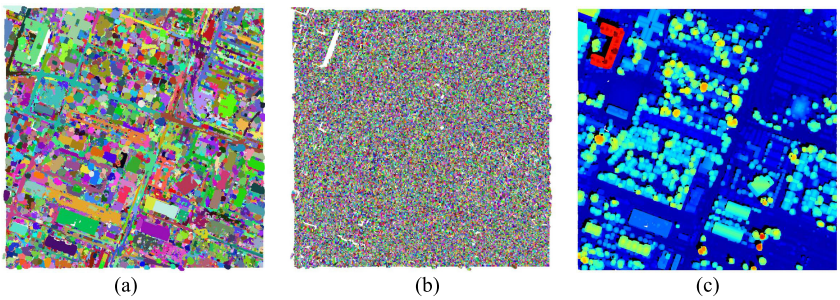


Fig. 11. Supervoxel segmentation and shadow detection results on real data. (a) Not regularized supervoxel segmentation. (b) Regularized supervoxel segmentation. (c) Shadow detection.

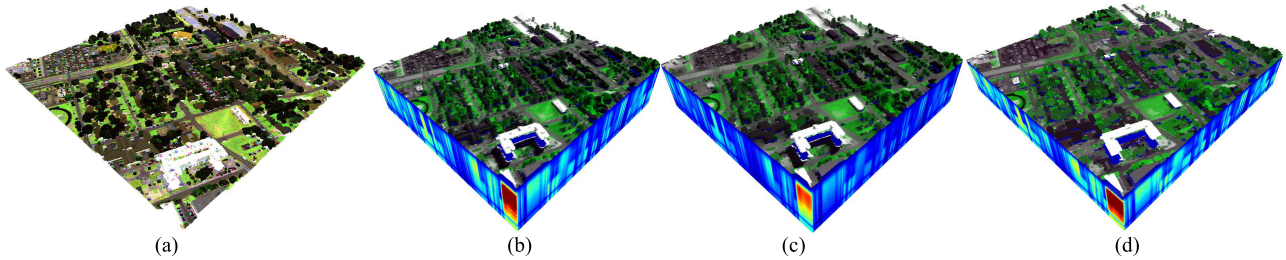
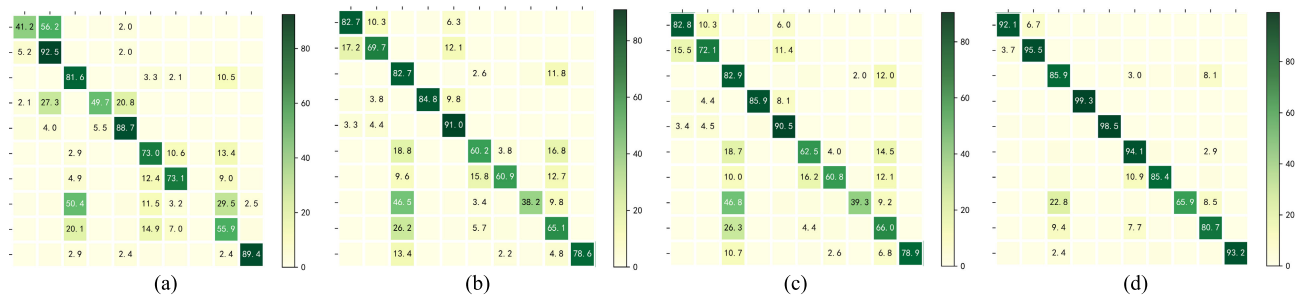


Fig. 12. Hyperspectral point cloud obtained by different methods on HU data. (a) MSPC. (b) HSPC-ori. (c) HSPC-2DRef. (d) Proposed.

shaded regions can be accurately classified as samples of other regions in the scene.

3) Parameter Influence and Complexity: The proposed supervoxel algorithm has two parameters, δ_μ and δ_I. δ_μ has effects on the degree of weighted graph mapped from the stereo hyperspectral data because the setting of δ_μ is related to the nominal spatial resolution of the stereo hyperspectral data and the density of point cloud. In this article, δ_μ is fixed at twice the spatial resolution of the stereo hyperspectral data, and each vertex in this weighted graph is connected to about 20 other adjacent vertices. δ_I is a more important parameter, and the size of it determines the scale of generated supervoxel.

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(a) (b) (c) (d)

Fig. 13. Confusion matrix obtained by different methods on samples from shadowed areas. (a) MSPC + Z(OA:74.42). (b) HSPC-ori + Z(OA:81.36). (c) HSPC-2DRef + Z(OA:81.68). (d) Proposed + Z(OA:90.24).

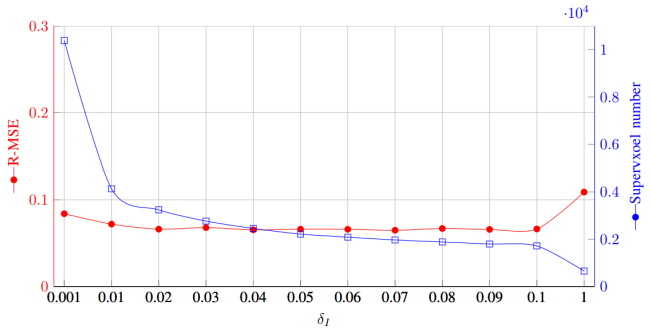


Fig. 14. Influence of the parameter δ_I on the proposed method.

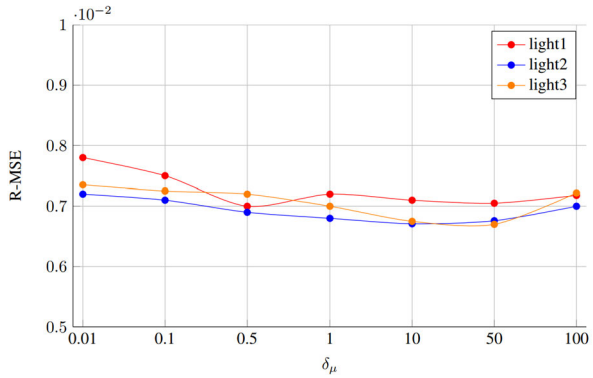


Fig. 15. Influence of the parameter δ_μ on the proposed method.

The scale of supervoxel can be measured by the number of eventually generated supervoxels, that is, the smaller the scale, the more supervoxels will be generated. Fig. 14 shows the effect of parameter δ_I on the proposed method that with the increase of δ_I, the number of supervoxels gradually decreases, while the RMSE error of the proposed method decreases at first and then increased. On one hand, the scale of supervoxel will be small when δ_I is too small, which leads to a severe oversegmentation situation where the information related to objects cannot be characterized well so that the RMSE error increases. On the other hand, if δ_I is too large, the scale of supervoxel will be large as well, and there will be a serious undersegmentation problem in turn, which introduces a segmentation error to the model.

The computational complexity analysis in point cloud generation can be divided into three parts: 1) in graph-based supervoxel segmentation processing, the computational complexity is related to the number of hyperspectral point clouds, which is $O(v)$; 2) in supervoxel-based shadow detection process, the computational complexity is related to the number of supervoxels, which is $O(t)$; and 3) in optimization part, the computational complexity is $O(v^2)$. Although the more supervoxels, the higher complexity of our algorithm, it is not hard to discover the fact that the RMSE error of δ_I does not change much within the interval [0.01, 0.1], and in this article, we fix δ_I at 0.05. Besides, the parameter δ_μ adjusts the weight coefficient between the imaging model term and the reflectance prior term, and its influence on the proposed method is drawn in Fig. 15; as you can see, with a large range of parameter δ_μ, RMSE changes are particularly small and almost negligible, which illustrates the robustness of our proposed approach.

V. Conclusion

In this article, we provide a novel hyperspectral point cloud IID model that can detect the shaded regions and recover reflectance information in them. From the perspective of imaging principle, there are three reasons for the uncertainties in the detection of hyperspectral remote sensing information: 1) the degeneration of spectral information; 2) hyperspectral 2-D imaging causes the degeneration of 3-D spatial information; and 3) the blocking among objects causes shadows in the data. To solve these problems, we first estimate the global illumination of the scene using an intrinsic information recovery method. Then, we perform supervoxel segmentation on hyperspectral point cloud, to calculate the blocking relation of supervoxels and therefore accurately detect shaded regions. Finally, we estimate the illumination and reflectance of shaded regions based on an illumination-invariant spectral prior. The experimental results show that this method can effectively detect shaded regions and robustly recover shadow-less IHSPC.

References

[1] C. Li, K. Zhou, and S. Lin, "Intrinsic face image decomposition with human face priors," in *Proc. Eur. Conf. Comput. Vis.*, D. Fleet, T. Pajdla, B. Schiele, and T. Tuytelaars, Eds. Cham, Switzerland: Springer, 2014, pp. 218–233.

[2] J. Shen, X. Yang, Y. Jia, and X. Li, "Intrinsic images using optimization," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2011, pp. 3481–3487.

[3] L. Shen and C. Yeo, "Intrinsic images decomposition using a local and global sparse representation of reflectance," in *Proc. CVPR*, Jun. 2011, pp. 697–704.

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[4] M. F. Tappen, W. T. Freeman, and E. H. Adelson, "Recovering intrinsic images from a single image," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 27, no. 9, pp. 1459–1472, Sep. 2005.

[5] L. Shen, C. Yeo, and B.-S. Hua, "Intrinsic image decomposition using a sparse representation of reflectance," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 12, pp. 2904–2915, Dec. 2013.

[6] J. T. Barron and J. Malik, "Shape, albedo, and illumination from a single image of an unknown object," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit., June*, vol. 2012, pp. 334–341.

[7] R. Basri, D. W. Jacobs, and I. Kemelmacher, "Photometric stereo with general, unknown lighting," *Int. J. Comput. Vis.*, vol. 72, no. 3, pp. 239–257, May 2007, doi: 10.1007/s11263-006-8815-7.

[8] S. Kim, K. Park, K. Sohn, and S. Lin, "Unified depth prediction and intrinsic image decomposition from a single image via joint convolutional neural fields," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2016, pp. 143–159.

[9] E. Shelhamer, J. T. Barron, and T. Darrell, "Scene intrinsics and depth from a single image," in *Proc. IEEE Int. Conf. Comput. Vis. Workshop (ICCVW)*, Dec. 2015, pp. 37–44.

[10] P.-Y. Laffont and J.-C. Bazin, "Intrinsic decomposition of image sequences from local temporal variations," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 433–441.

[11] Z. Li and N. Snavely, "Learning intrinsic image decomposition from watching the world," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 9039–9048.

[12] Q. Chen and V. Koltun, "A simple model for intrinsic image decomposition with depth cues," in *Proc. IEEE Int. Conf. Comput. Vis.*, Dec. 2013, pp. 241–248.

[13] M. Hachama, B. Ghanem, and P. Wonka, "Intrinsic scene decomposition from RGB-D images," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 810–818.

[14] J. Jeon, S. Cho, X. Tong, and S. Lee, "Intrinsic image decomposition using structure-texture separation and surface normals," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2014, pp. 218–233.

[15] N. Kong and M. J. Black, "Intrinsic depth: Improving depth transfer with intrinsic images," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1–9.

[16] J. T. Barron and J. Malik, "High-frequency shape and albedo from shading using natural image statistics," in *Proc. CVPR*, Jun. 2011, pp. 2521–2528.

[17] Y. Gu, W. Xie, X. Li, and X. Jin, "Hyperspectral intrinsic image decomposition with enhanced spatial information," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5523014.

[18] X. Jin and Y. Gu, "Superpixel-based intrinsic image decomposition of hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 8, pp. 4285–4295, Aug. 2017.

[19] X. Jin, Y. Gu, and T. Liu, "Intrinsic image recovery from remote sensing hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 224–238, Aug. 2019.

[20] X. Kang, S. Li, and J. A. Benediktsson, "Intrinsic image decomposition for feature extraction of hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 5, pp. 2666–2677, May 2015.

[21] T. Hakala, J. Suomalainen, S. Kaasalainen, and Y. Chen, "Full waveform hyperspectral LiDAR for terrestrial laser scanning," *Opt. Exp.*, vol. 20, no. 7, pp. 7119–7127, Mar. 2012.

[22] E. Puttonen et al., "Artificial target detection with a hyperspectral LiDAR over 26-h measurement," *Opt. Eng.*, vol. 54, no. 1, 2015, Art. no. 013105.

[23] J. Vauhkonen et al., "Classification of spruce and pine trees using active hyperspectral LiDAR," *IEEE Geosci. Remote Sens. Lett.*, vol. 10, no. 5, pp. 1138–1141, Sep. 2013.

[24] L. Du et al., "Estimation of rice leaf nitrogen contents based on hyperspectral LiDAR," *Int. J. Appl. Earth Observ. Geoinformation*, vol. 44, pp. 136–143, Feb. 2016.

[25] J. C. Fernandez-Diaz et al., "Capability assessment and performance metrics for the Titan multispectral mapping LiDAR," *Remote Sens.*, vol. 8, no. 11, p. 936, 2016.

[26] M. Khodadadzadeh, J. Li, S. Prasad, and A. Plaza, "Fusion of hyperspectral and LiDAR remote sensing data using multiple feature learning," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2971–2983, Jun. 2015.

[27] W. Liao, A. Pizurica, R. Bellens, S. Gautama, and W. Philips, "Generalized graph-based fusion of hyperspectral and LiDAR data using morphological features," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 3, pp. 552–556, Mar. 2015.

[28] P. Ghamisi et al., "Multisource and multitemporal data fusion in remote sensing: A comprehensive review of the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 1, pp. 6–39, Mar. 2019.

[29] Q. Guo et al., "Lidar boosts 3D ecological observations and modelings: A review and perspective," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 1, pp. 232–257, Mar. 2021.

[30] Y. Gu, Q. Wang, X. Jia, and J. A. Benediktsson, "A novel MKL model of integrating LiDAR data and MSI for urban area classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 10, pp. 5312–5326, Oct. 2015.

[31] P. Ghamisi and B. Höfle, "LiDAR data classification using extinction profiles and a composite kernel support vector machine," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 5, pp. 659–663, May 2017.

[32] Y. Gu and Q. Wang, "Discriminative graph-based fusion of HSI and LiDAR data for urban area classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 6, pp. 906–910, Jun. 2017.

[33] Q. Wang and Y. Gu, "A discriminative tensor representation model for feature extraction and classification of multispectral LiDAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 3, pp. 1568–1586, Mar. 2019.

[34] Q. Man, P. Dong, and H. Guo, "Pixel-and feature-level fusion of hyperspectral and LiDAR data for urban land-use classification," *Int. J. Remote Sens.*, vol. 36, no. 6, pp. 1618–1644, 2015.

[35] L. Liu, N. C. Coops, N. W. Aven, and Y. Pang, "Mapping urban tree species using integrated airborne hyperspectral and LiDAR remote sensing data," *Remote Sens. Environ.*, vol. 200, pp. 170–182, Oct. 2017.

[36] S. J. Buckley, T. H. Kurz, J. A. Howell, and D. Schneider, "Terrestrial LiDAR and hyperspectral data fusion products for geological outcrop analysis," *Comput. Geosci.*, vol. 54, pp. 249–258, Mar. 2013.

[37] T. Sankey, J. Donager, J. McVay, and J. B. Sankey, "UAV LiDAR and hyperspectral fusion for forest monitoring in the southwestern USA," *Remote Sens. Environ.*, vol. 195, pp. 30–43, Jun. 2017.

[38] Y. Cui et al., "Deep learning for image and point cloud fusion in autonomous driving: A review," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 2, pp. 722–739, Feb. 2022.

[39] H. Su, S. Maji, E. Kalogerakis, and E. Learned-Miller, "Multi-view convolutional neural networks for 3D shape recognition," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 945–953.

[40] R. Klokov and V. Lempitsky, "Escape from cells: Deep Kd-networks for the recognition of 3D point cloud models," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 863–872.

[41] J. M. Rubin and W. A. Richards, "Color vision and image intensities: When are changes material?" *Biol. Cybern.*, vol. 45, no. 3, pp. 215–226, Oct. 1982.

[42] G. Healey and D. Slater, "Models and methods for automated material identification in hyperspectral imagery acquired under unknown illumination and atmospheric conditions," *IEEE Trans. Geosci. Remote Sens.*, vol. 37, no. 6, pp. 2706–2717, Nov. 1999.

[43] X. Jin, Y. Gu, T. Liu, and W. Xie, "Supervoxel-based intrinsic scene properties from hyperspectral images and LiDAR," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2022.

[44] Y. Xu et al., "Advanced multi-sensor optical remote sensing for urban land use and land cover classification: Outcome of the 2018 IEEE GRSS data fusion contest," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 6, pp. 1709–1724, Jun. 2019.

[45] S. Luo, H. Li, R. Zhu, Y. Gong, and H. Shen, "ESPFNet: An edge-aware spatial pyramid fusion network for salient shadow detection in aerial remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 4633–4646, 2021.

[46] H. Le and D. Samaras, "Shadow removal via shadow image decomposition," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 8577–8586.

[47] E. Arbel and H. Hel-Or, "Shadow removal using intensity surfaces and texture anchor points," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 33, no. 6, pp. 1202–1216, Jun. 2011.

[48] Z. Ling, Q. Zhang, and C. Xiao, "Shadow remover: Image shadow removal based on illumination recovering optimization," *IEEE Trans. Image Process.*, vol. 24, no. 11, pp. 4623–4636, Nov. 2015.

[49] Z. Liu, H. Yin, X. Wu, Z. Wu, Y. Mi, and S. Wang, "From shadow generation to shadow removal," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 4925–4934.

[50] P. Duan, S. Hu, X. Kang, and S. Li, "Shadow removal of hyperspectral remote sensing images with multiexposure fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–11, 2022.

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[51] P. Hartzell, C. Glennie, and S. Khan, "Terrestrial hyperspectral image shadow restoration through LiDAR fusion," *Remote Sens.*, vol. 9, no. 5, p. 421, Apr. 2017.

[52] X. Yu, G. Li, Z. Ying, and X. Guo, "A new shadow removal method using color-lines," in *Proc. Int. Conf. Comput. Anal. Images Patterns*. Cham, Switzerland: Springer, 2017, pp. 307–319.



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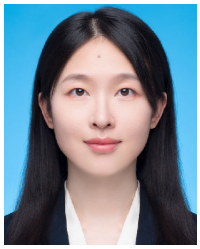
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